# Project INSPIRE Intro Meeting

Liam Gallagher

# Agenda

- Introduction
- Background/Vision
- Objectives / Scope of Work
- Team Roles & Responsibilities
- Information Gathering & Q&A



# So Who Am I?

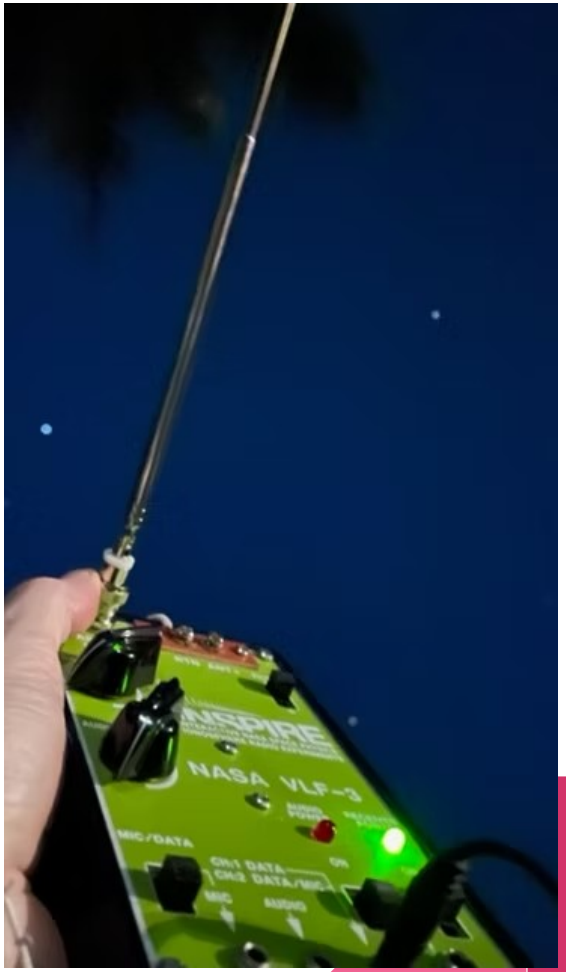
- Liam Gallagher, BSEE Student
- Undergraduate Researcher at Florida Space Institute
- Radio Astronomy Lab
  - RF Signal Chain Design
  - FPGA System Development
  - Signal Processing Algorithm Development
  - Systems Integration & Testing





# Vision

The goal of this project is to build a modern incarnation of the NASA INSPIRE VLF-3 Radio Receiver.

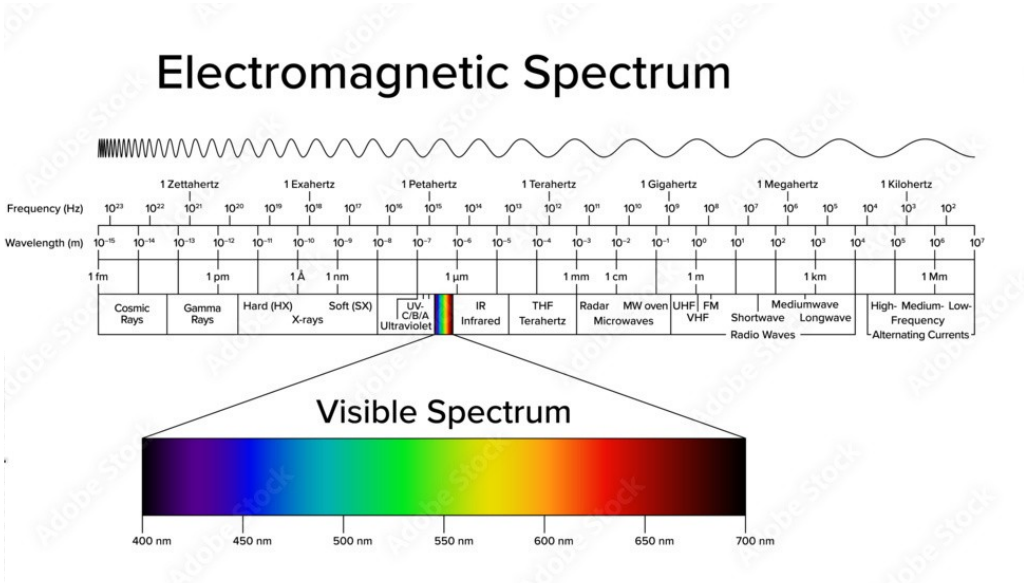


# Scientific Background

Electromagnetic Waves include Radio, Infrared, Visible, UV, and X-Rays

Fundamental Frequency-Wavelength Relationship is:

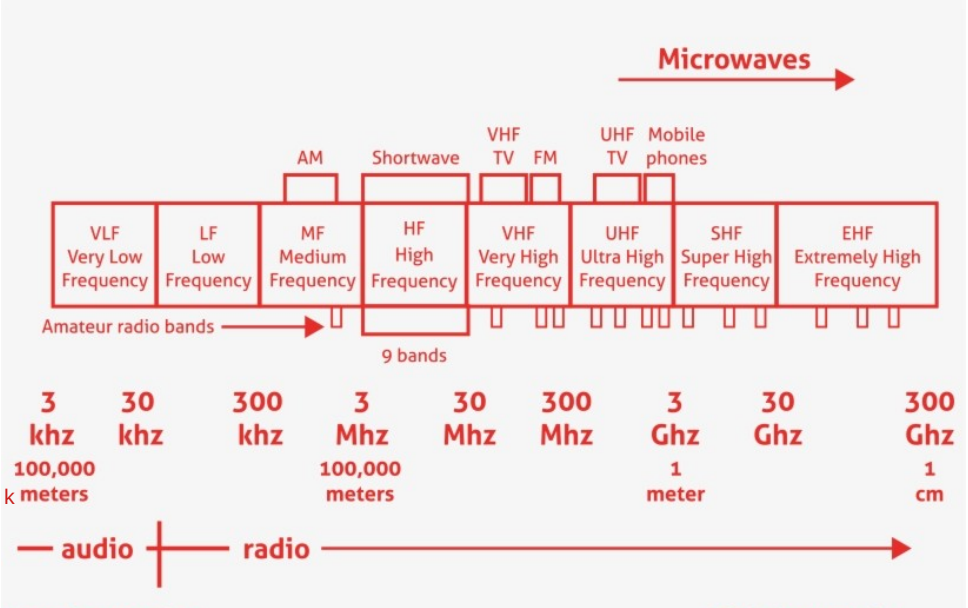
$$\lambda = \frac{c}{f}$$



# Scientific Background II

As you go up in Radio Frequency, Cost and Complexity Increase FAST

As such, we will be doing our RF Project in the lowest Frequency, VLF (Very Low Frequency)

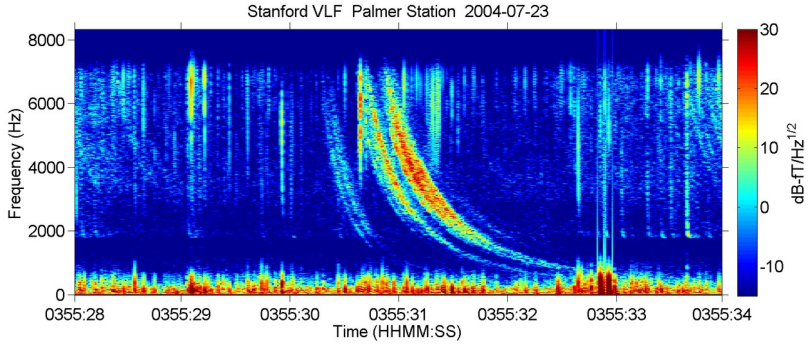


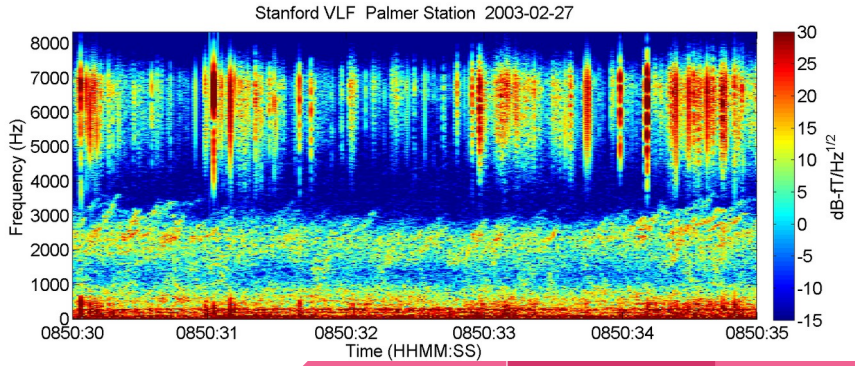
# Scientific Background III

Low Frequency ≠ Not Cool

Things in VLF range include Submarine Communications, Long-Range Navigation (LORAN), and Natural Phenomena

- Lightning Strikes (top right)
- Ionospheric Entries
- Dawn Chorus (bottom right)



# NASA INSPIRE VLF-3

- Receiver for VLF waves; amplifies and filters antenna input to be sent to headphones/cassette tape
  - 1980s/1990s Design
  - ~30dB of gain
  - Single Antenna

- We want to make our own design



# High Level Design

- Three Main Parts
  - Antenna (off-the shelf)
  - Front-End (Analog Circuitry)
  - Back-End (Digital Recording)
- INSPIRE VLF-3 only has the middle
- Back-end will be a MCU-powered recorder

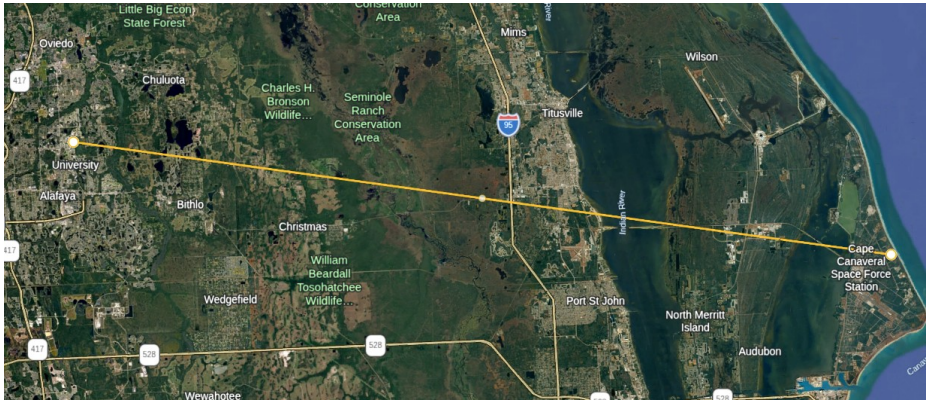


# Overall Design Specifications

- Match or Beat VLF-3
  - 30 dB of Gain
  - 3-30 kHz Bandwidth
  - High Input Impedance
  - Single Antenna
- Modular PCB Construction
  - Dual PCB allows for separation of Front-End and Back-End
  - Easier to Test & Fix/Replace, Less Noise Leakage
- Digital Component Integrated
  - No need to bring Laptop/PC
  - Makes Field Testing Easier

# Why No Dual Antenna?

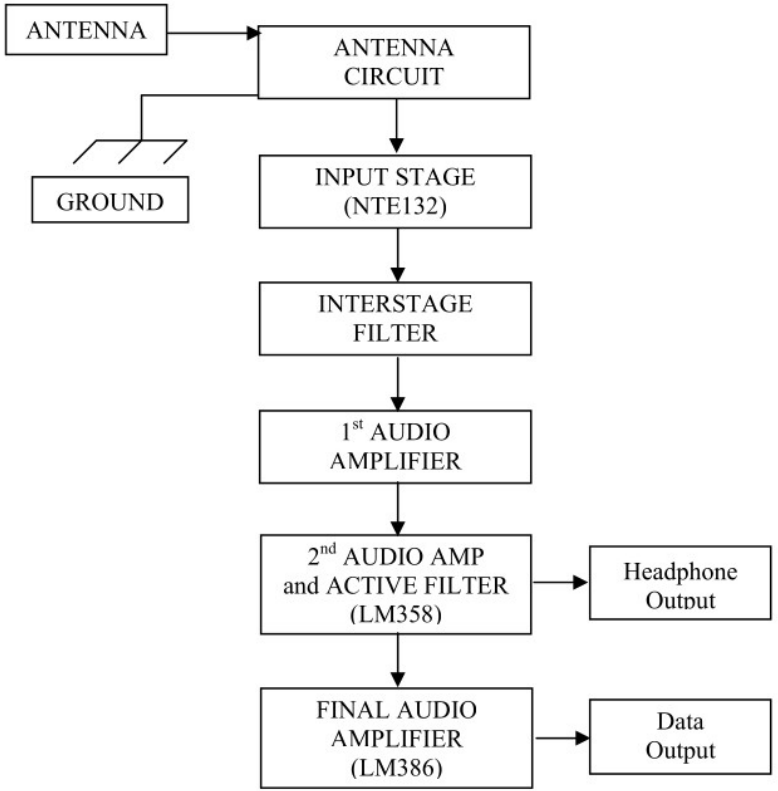
- Dual Antennas let us determine incoming direction of waves
  - At VLF, our antennas would require miles of separation and perfect synchronization.
  - Not possible for VLF without outside help.




UCF to KSC is less then a wavelength!

# Team I: Analog

- Responsible for design and build of Front-End
  - Right is VLF-3 Front-End Design
  - Modern Silicon Advances means this can be simplified
- Each block can be designed and prototyped separately
- Analog Lead responsible for overseeing this design
  - I can provide mini-workshops on this design



# Team II: Digital

- Responsible for design and software of Back-End
  - Stereo Audio Recorder
  - Dual ADC + MCU + SD Card
- Once Hardware is designed and built, optional implementation of DSP
- Digital Lead will need to coordinate hardware and software
  - Possible for fewer digital hardware designers & emphasis on software/DSP
  - Assistance will be provided if needed



# Information Gathering

- Open-ended Project
  - I’d like to get a feel for the people on the teams’ skill/experience/confidence
- QR Code to Survey
  - If you haven’t filled out already from introduction announcement, please use the QR code on the right, or the link in #inspire-announcements

# Thank you for Listening!

Any Questions?

- Can be about RF or this Project